Round Robin CPU Scheduling

Computer Science Learning Module | Participant Reading Material

Learning goals

After this lesson, you should be able to explain Round Robin scheduling, describe the role of the time quantum and ready queue, trace a simple schedule, and calculate waiting and turnaround times.

1. What is Round Robin?

Round Robin (RR) is a preemptive CPU scheduling algorithm. Processes wait in a ready queue, and the CPU gives each process up to a fixed amount of time called the **time quantum**. If a process finishes early, it leaves the queue. If it still has CPU work remaining when its quantum expires, the scheduler preempts it and places it at the back of the ready queue.

2. How the algorithm works

1. Place ready processes in a first-in, first-out (FIFO) queue.

2. Run the process at the front for at most one time quantum.

3. If it finishes, remove it; otherwise, put it at the back of the queue.

4. Repeat until every process completes.

3. Worked example

Assume all three processes arrive at time 0 and the time quantum is **2 ms**.

Process	Arrival time	CPU burst time
P1	0 ms	5 ms
P2	0 ms	3 ms
P3	0 ms	1 ms

Timeline: 0–2 P1 | 2–4 P2 | 4–5 P3 | 5–7 P1 | 7–8 P2 | 8–9 P1

P1 first runs for 2 ms and has 3 ms left. P2 runs for 2 ms and has 1 ms left. P3 finishes in 1 ms. P1 runs again for 2 ms, then P2 finishes its last 1 ms, and finally P1 finishes its last 1 ms.

4. Completion, turnaround, and waiting times

Completion time (CT) is when a process finishes. **Turnaround time (TAT)** = completion time – arrival time. **Waiting time (WT)** = turnaround time – CPU burst time (assuming no I/O blocking).

Process	CT	TAT	WT
P1	9	9 – 0 = 9	9 – 5 = 4
P2	8	8 – 0 = 8	8 – 3 = 5
P3	5	5 – 0 = 5	5 – 1 = 4

Average waiting time = (4 + 5 + 4) / 3 = **4.33 ms**. **Average turnaround time** = (9 + 8 + 5) / 3 = **7.33 ms**.

5. Why the time quantum matters

A very small time quantum gives processes frequent turns but increases context-switch overhead. A very large time quantum reduces switching but can make Round Robin behave more like First-Come, First-Served (FCFS). Choosing an appropriate quantum balances responsiveness and overhead.

6. Strengths and limitations

Strengths: fair CPU access for ready processes; useful responsiveness for time-sharing systems; no single CPU-bound process monopolizes the CPU. **Limitations:** context switches have overhead; performance depends on quantum size; average waiting time is not always optimal.

7. Quick recap

Round Robin repeatedly gives each ready process a turn of up to one time quantum. Unfinished processes return to the back of the ready queue. Use the timeline to find completion times, then calculate turnaround and waiting times.

Study instruction: Please review the material carefully before beginning the quiz. The quiz will have 10 questions and a 15-minute time limit.